

grapple with concerns about biocompatibility—that is, making sure that the materials are not toxic and do not trigger harmful immune reactions. Hopefully these challenges could be surmounted and that nanotechnology would help bring about a golden era of diagnostics and medicine.

Clean Water

In first world countries, the potential application of Nanotechnology focuses more on energy and medicine, but in the third-world countries, the focus shifts to water resources. After all it is an essential part of life and is crucial to our standard of living, greatly affecting our ability to maintain health, grow food and build vibrant industries. Nanotechnology has the potential to revolutionize clean-water technologies by improving water purification, preventing water pollution and cleaning up tainted groundwater, lakes and streams.

According to the United Nations, one-fifth of the world's population lacks access to safe drinking water. Climate change is expected to further strain water resources by shifting precipitation patterns and causing severe drought in some regions. Furthermore this problem is not restricted to the under-developed and developing countries only; much of the central United States is experiencing record-breaking drought. Water woes even plague communities with plentiful supplies, but poor water quality. Old pipes leaching unsafe amounts of lead into the water, groundwater contaminated by perchlorate and other contaminants, render them unfit for use.

Nanotechnology can be exploited to boost the availability of clean water. Nanotechnology could be used to determine water cleanliness, enable the filtration of dirty water, reduction of water use, and cleaning up polluted bodies of water. Nanomaterials could be harnessed to enhance existing water-purification processes as they have a high surface area and can be chemically tailored. Nanoparticles show great potential as sorbents, materials that latch on to pollutants and pull them out of solution. For instance, multi-walled carbon nanotubes have been shown to take up lead, cadmium and copper more effectively than does activated carbon, a commonly used sorbent. Some nanoparticles also act



The need of the Hour.